Mapping 102 autism-risk genes in relation to the Sodium-Struggle Hypothesis: a gated phenotype-first systems-physiology classification

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Abstract

Background: The Sodium-Struggle Hypothesis proposes that recurrent instability across sodium-weighted extracellular support, effective circulating volume, perfusion and cellular ion recovery may shape development. Genetic variation is not required, but variants may alter susceptibility, physiological load, controller execution, reserve or recovery. The fixed Satterstrom-102 set provides an independent rare-variant gene set for examining this proposed systems architecture.

Objective: To map the fixed 102 autosomal autism-risk genes within a version-controlled phenotype-first systems-physiology classification and determine which gene-specific endpoints meet a prespecified final evidence gate.

Methods: Genes were first mapped within hypothesis-derived physiological buckets and then re-reviewed individually in alphabetical order with category assignment suspended until the measured phenotype, direction, allele or dosage, compartment, developmental window and challenge state had been established. Final rows underwent whole-map harmonization, ascertainment control, retrospective baseline audit and numerator freeze.

Results: Forty-two genes cleared one qualifying gate across ten heterogeneous endpoint families. Fifty-six Partial rows were retained as informative non-counting holds: 35 primary placements in support or contextual layers that were non-counting by construction, 16 allele, dosage or model mismatches, and five missing functional, challenge-state or systemic bridges. Partial therefore means that informative biology could not legitimately enter the strict numerator under the frozen rules; it does not mean that the gene was tested against the framework and failed. Two genes were explicit boundary exclusions, one remained unresolved and one was a bounded cross-reference. The complete map identified routes involving ionic and chloride load, endocrine and secretory control, vascular and lymphatic containment, cardiac output and respiratory control, structural and energetic reserve, and target-tissue recovery. Category size described occupancy, not effect size or physiological leverage.

Conclusions: The fixed gene set maps onto a distributed, predominantly upstream physiological architecture relevant to the Sodium-Struggle Hypothesis without implying that every gene controls serum sodium or circulating volume. This descriptive classification is not an enrichment test and does not establish a shared sodium-volume state in autistic people. Its main outputs are a transparent map, a stratified account of why informative genes remained non-counting, an auditable frozen 42/102 numerator and a falsifiable experimental programme.

Keywords: autism spectrum disorder; autism genetics; Sodium-Struggle Hypothesis; systems physiology; phenotype-first classification; physiological reserve; challenge-state physiology; evidence mapping; systems-physiology mapping

1. Introduction

1.1 The interpretive problem created by large autism-gene sets

Large-scale exome sequencing has produced gene sets of a size and confidence that were unavailable a decade ago. Satterstrom and colleagues identified 102 autosomal genes at a false discovery rate below 0.10, including many genes not detected in earlier exome analyses and many not previously implicated in autosomal dominant neurodevelopmental disorders [1]. The genes do not carry equal statistical support, and even the best-supported individual genes account for only a small fraction of autistic people.

The difficulty is no longer identifying genes. It is explaining how rare variants in a strikingly heterogeneous assortment of genes can contribute to overlapping autistic phenotypes. The source study emphasized gene-expression regulation, neuronal communication and cytoskeletal organization [1]. Those categories are informative, but they do not specify what the genes change at the level of organs, physiological control systems, challenge responses or recovery.

The Satterstrom-102 set is one high-confidence rare-variant window into autism, not a genetic account of autism as a whole. Many autistic people carry no identifiable high-impact de novo coding variant; inherited rare variation and common polygenic variation also contribute substantially to risk [2,7,8]. The rarity of each individual gene association makes the convergence question more, rather than less, important: the unresolved issue is the level at which many different genetic routes can lead toward a related neurodevelopmental diagnosis.

1.2 What convergence analyses have examined

The 102 genes and related high-confidence autism genes have been grouped in several influential ways. Satterstrom and colleagues separated genes broadly into regulation of gene expression and neuronal communication, contrasted genes relatively enriched in autism with those relatively enriched in severe neurodevelopmental delay, and localized expression to both excitatory and inhibitory cortical lineages [1]. Ruzzo and colleagues integrated inherited and de novo risk into shared biological networks [2]. Jin and colleagues perturbed 35 ASD/neurodevelopmental-delay genes in the developing mouse brain and identified recurrent neuronal and glial modules [3]. Paulsen and colleagues found that KMT5B, ARID1B and CHD8 haploinsufficiency converged on asynchronous development of GABAergic and deep-layer excitatory neurons in human organoids [4]. Li and colleagues screened 36 high-risk transcription-regulatory genes in cerebral organoids and identified vulnerable progenitor and neuronal states [5]. Gordon and colleagues later showed mutation-specific early effects that converged on shared transcriptional changes as development progressed across eight ASD-associated mutations [6]. These studies illuminate molecular, developmental and neural-cellular convergence; the present study asks whether the same fixed gene set also shows a coherent whole-organism physiological organization.

1.3 The missing level

Autism is diagnosed through behaviour and is usually investigated as a neurological condition. Yet for many genes in the fixed set, the most informative measured phenotype is not confined to neurons: it may involve placenta, vasculature, lymphatic vessels, heart, kidney, endocrine pancreas, gastrointestinal or enteric systems, immune cells, astrocytes, skeletal muscle or general cellular-

recovery machinery. This statement rests on measured organ, tissue or cellular findings, not on nonspecific outward signs such as hypotonia, constipation or abnormal gait, which were recorded separately and were not allowed to generate a physiological route.

These extracerebral phenotypes may be downstream comorbidities, parallel consequences of the same variant, or upstream contributors to the physiological conditions under which neurodevelopment unfolds. The classification was designed not to assume that a peripheral finding is merely a consequence of severe neurodevelopmental disease, but to ask what each gene changes in physiology: what challenge or load it creates, what sensing or controller process it alters, what reserve or deployment capacity it constrains, and how vulnerable tissues recover when support is imperfect.

1.4 What this study does

The analytical process had two deliberately different phases. The initial map examined genes within physiological buckets derived from the Sodium-Struggle Hypothesis. This generated many productive connections but also created a risk of bucket-first confirmation. The complete gene set was therefore re-examined one gene at a time in alphabetical order, with the category structure temporarily suspended. Reviewers first established the measured human or experimental phenotype, direction, model and dosage; only then was the gene reassigned to a primary physiological domain and given bounded secondary and creative routes. The final map reflects that second phenotype-first re-review plus whole-map harmonization, not the original bucket placement alone.

1.5 Disclosure of motivation

The named motivating framework is the Sodium-Struggle Hypothesis of Autism. Bell conceived its core premise in 2015 and developed it privately over the following decade; the current hypothesis is substantially more developed than the original 2015 formulation. The earliest surviving public posting is Zenodo record 19700479, version v5, published 19 October 2025 [9]. No claim is made that the 2015 formulation was identical to the present framework or independently time-stamped before that surviving posting. The core premise nevertheless predates, and was developed independently of, the 2020 Satterstrom-102 study. Genetic variation is not a necessary premise of the hypothesis. The hypothesis proposes a physiological process that may arise through genetic, developmental, maternal, environmental or acquired routes; particular variants may increase susceptibility or alter the cost, recruitment, execution and termination of defence.

Throughout this manuscript, sodium-volume-perfusion defence refers to a coupled whole-organism architecture rather than to renal sodium transport alone. It includes sodium-water and extracellular-tonicity regulation; effective central filling, venous return, cardiac output, vascular tone, capacitance, containment and regional distribution; renal, endocrine, cardiac, vascular and autonomic sensing and controller recruitment; tissue perfusion; and cellular ion, osmolyte and energetic execution and recovery. The narrower phrase canonical renal/RAAS/AVP or transporter-level sodium-handling machinery is reserved for the traditional nephron, renin-angiotensin-aldosterone, vasopressin and direct sodium-water transport branches.

These component systems are established physiology. The novel claim is that recurrent disturbance and incomplete reset across them may converge in a subset of autism. The analysis was not blind to that claim. The externally fixed gene set was specifically the 102 autosomal genes reported by Satterstrom and colleagues, and it was never modified on the basis of fit. The hypothesis manuscript and the analytical classifier were treated as distinct objects: the former supplied the motivating physiological model and was not rewritten in response to the gene-review results during the review series, whereas

the latter was iteratively refined, versioned and retrospectively harmonized when recurrent analytical failure modes were identified.

1.6 Objectives

1. Determine where the fixed Satterstrom-102 genes map within a systems-physiology ontology derived from, but not limited to, the Sodium-Struggle Hypothesis.
2. Determine which gene-specific endpoints meet the frozen final evidence gate.
3. Characterize the non-exclusive demand, controller or execution, and target-tolerance or recovery roles occupied across the complete set.
4. Identify, for non-counted genes, the specific measurement or model bridge whose absence prevented adjudication.
5. Document the conditions under which the classifier refuses or revokes an attractive connection.

The study evaluates correspondence between the fixed gene set and the hypothesis-derived physiological architecture, but correspondence is not treated as proof that the hypothesis is correct or exclusive.

2. Methods

2.1 Fixed gene set

The denominator was the fixed set of 102 autosomal genes identified by Satterstrom and colleagues at TADA false discovery rate <= 0.10 [1]. Bell had no role in creating or selecting this gene set. No gene was added, removed or substituted on the basis of physiological fit. Source symbols were retained where they differ from current nomenclature, with current symbols recorded separately in the workbook.

The final analytical state was frozen in workbook version v46.20. The active authority comprised the Final 102-Gene Map, Count-by-Gate register, 33-domain summary, completed 102-row Systems Role Matrix, completed 102-row ASD Ascertainment-to-Evidence Matrix, retrospective status and baseline audits, and numerator-freeze ledger [10-12].

2.2 Physiological ontology and publication umbrellas

The technical ontology contains 33 physiological, supporting and contextual bins. Twenty-four are count-eligible physiological architecture domains, in which a sufficiently direct gene-specific endpoint may qualify. Domains 25, 26, 27 and 30 are supporting or downstream layers that are non-counting by membership unless a separate direct endpoint closes. Domains 28 and 29 are exploratory boundaries, Domains 31 and 32 are contextual classes, and Domain 33 is unresolved or pre-mechanistic. What unifies the bins is their position along the hypothesis's proposed causal architecture: challenge or demand generation; sensing and defence execution; structural, energetic or secretory reserve; and target-tissue tolerance or recovery.

For main-text readability, the 33 technical domains were mapped once and only once into seven publication umbrellas. This presentation layer was created after the technical ontology was frozen and has no count effect. The seven-umbrella mapping used in this manuscript is the canonical presentation mapping and supersedes earlier draft-level umbrella schemes; it was selected because each technical domain is assigned exactly once and the totals reconcile to the fixed denominator of 102. The umbrellas are not assumed to have equal causal weight. They summarize different ways in which a gene could alter the probability, cost or developmental consequence of Sodium-Struggle; the complete technical mapping and qualifying endpoint for each domain belong in Supplementary Table S2. Publication umbrellas are deliberately broader than qualifying gate families: for example, Figure 2 combines chloride/inhibitory maturation with direct ionic excitability, whereas Table 3 separates those gate families because their qualifying endpoints differ.

Table 1. Publication umbrellas used in the main text [10]. The umbrellas describe different positions within one coupled sodium-volume-perfusion defence architecture; they are not ranked by distance from a renal core. The label support denotes a functional position, not physiology external to that architecture. The final column states the proposed position of each umbrella and does not establish that every member causes the hypothesized physiological state. Strict counts are not pass rates for structurally non-counting layers.

Publication umbrella	Technical domains	Genes	Strict counts	Physiological content and relation to Sodium-Struggle
Structural, membrane and extracellular containment	Domains 1, 7	5	3	Boundary integrity and containment. These genes can alter whether cells, axons, receptor microdomains or extracellular/lymphatic compartments remain stable during osmotic, perfusion or mechanical stress.
Astrocytic, osmotic, CSF and energetic recovery	Domains 2, 5, 6, 24	9	7	Local ion and water clearance, mitochondrial capacity and glial/CSF reset. These routes modify the ability to recover after neural or systemic challenge.
Chloride, inhibitory maturation and ionic excitability	Domains 3, 4, 26	20	15	Ionic workload, chloride persistence and inhibitory maturation. These routes can increase the energy cost of restoring gradients and prolong immature or unstable signalling states.
Renal, mineral and endocrine control	Domains 8-14, 17	7	3	Renal, mineral and endocrine sensing, retention and peptide set points. This umbrella contains the narrow canonical renal/mineral/endocrine controller branch; it is one component of the broader sodium-volume-perfusion defence architecture.
Secretory, circadian and metabolic response	Domains 15, 16, 18, 27, 28	15	6	Hormone or neurotransmitter deployment, circadian timing and metabolic substrate support. These routes modify controller, effector and recovery execution within the same broader architecture.
Placental, vascular, cardiac, autonomic and mechanical support	Domains 19-23, 29	13	8	Placental delivery, vascular or lymphatic containment, cardiac output, respiratory gain, autonomic rescue and mechanical contributions to filling and flow. These routes occupy circulatory and controller positions within the same broader sodium-volume-perfusion defence architecture.
Immune, epithelial, cellular-recovery and contextual developmental layers	Domains 25, 30-33	33	N/A	Illness demand, epithelial/barrier stress, general cellular recovery and contextual development. These layers can amplify challenge or tissue vulnerability but are non-counting by primary membership.

The umbrella counts describe where genes were placed, not how powerful any one gene or category is. A singleton controller may exert more physiological leverage than a large collection of weak modifiers,

whereas a large category may partly reflect what the literature has measured. Counts therefore indicate occupancy and gate output, not effect size, prevalence or causal importance.

2.3 Chronology of the ontology and meaning of freeze

1. The 102-gene denominator was fixed externally before the analysis and never altered.
2. The core premise was conceived in 2015 and developed privately over the following decade; the current hypothesis is more developed than the original 2015 formulation. The earliest surviving public posting is Zenodo version v5, published 19 October 2025 [9], and no claim of independent time-stamping before that date is made. The hypothesis manuscript was not revised in response to the gene-review results during the review series.
3. The domain ontology, evidence ladder and adjudication rules were refined during early review passes, principally in response to identified failure modes.
4. All modifications were versioned, with input and output provenance recorded.
5. Retrospective sweeps brought genes reviewed under earlier rules into conformity with later rules.
6. The ontology, status conventions, ascertainment controls and strict numerator were frozen before manuscript analysis under freeze identifier NF-2026-08-11-v46.20.

Any later numerator change must be recorded as an explicit post-freeze amendment. Silent edits are prohibited.

2.4 Two-stage review design and phenotype-first protocol

The first analytical phase mapped genes within hypothesis-derived physiological buckets. The second phase re-reviewed all genes individually in alphabetical order, with categories suspended until the measured phenotype had been established. The final domain assignment was made after this category-suspended review. The operational sequence for the second phase was:

1. Identify the human disease allele class or the most disease-relevant experimental perturbation available.
2. Determine what measurably fails in the organism, organ, tissue or cell.
3. Record the measured endpoint and direction.
4. Separate initiating pathology from successful compensation, failed compensation and downstream cost.
5. Only then map the phenotype to the physiological ontology.

During the alphabetical re-review, the analysis did not begin by asking how a gene could fit a preassigned bucket. The Sodium-Struggle lens still shaped the questions asked and the later systems interpretation; the category-suspended step was introduced to reduce, not eliminate, framework-driven confirmation bias.

2.5 Model interpretation: knockout-first but not knockout-only

Complete and conditional knockout models were used to discover organismal function, but not treated as automatically disease-equivalent. Homozygous null was separated from human heterozygosity, tissue-specific deletion from germline disease, gain of function from loss of function, and severe or lethal phenotypes from primary gene function. Placental, vascular and gravitational physiology were not assumed to transfer directly from mouse to human. Compensatory replacement of a deleted protein by a paralogue or alternate product was treated as a model-integrity problem rather than clean loss of function.

2.6 Evidence ladder

Level	Operational meaning
Level 1	Directly measured gene-specific phenotype.
Level 2	Physiological interpretation following closely from the measured phenotype.

Level	Operational meaning
Level 3	Stepwise-supported systems connection with one or more unmeasured joints.
Level 4	Mechanistically coherent, explicitly testable hypothesis.
Rejected	Inference violates dosage, direction, compartment, specificity or evidence-type rules.

Level 4 material was preserved as a separate non-counting discovery layer, with the decisive experiment attached. Rejected routes were retained when they illustrated an important boundary or failure mode.

2.7 Variant x Compartment x Developmental Window x State

No gene was assigned a single universal direction. Interpretation was conditioned on allele class, tissue or cell compartment, developmental window and physiological state. The same channel gene may contain gain- and loss-of-function disease branches; a perturbation may increase excitability in one cell type and reduce it in another; an endothelial phenotype cannot be assigned to neurons; and a challenge deficit may coexist with a relatively adequate resting average.

2.8 Specificity attribution

Attribution class	Meaning
Disease-gene-specific phenotype	Measured for the gene at disease-relevant dosage or allele.
Exact-protein mechanism	Property of the normal protein or exact protein, not necessarily of the disease allele.
Complex or assembly property	Produced by disruption or reconstitution of a multi-protein system.
Family or paralogue evidence	Shared or borrowed from a related gene; never counted as gene-specific.
Cell-population marker	Property of cells the gene labels, not proof that the gene executes that function.

The control question was whether a disease-irrelevant member of the same class, complex or family would be expected to show the same endpoint. If so, the finding was labelled as a class or assembly effect. Marker-positive cells were not treated as evidence of gene function unless the gene itself was perturbed in that compartment.

2.9 Ascertainment allele-class control

For every gene, the final matrix recorded: the ASD-ascertained allele class; the allele or model class carrying the physiological evidence; the relationship between them; and whether direction was measured, inferred or unresolved. Exact-variant match, same-class match, partial match, different-class evidence and indirect evidence were distinguished.

This control was count-sensitive. DYNC1H1 lost a previously granted count when qualifying mitochondrial evidence was found to derive from missense models while the ASD ascertainment was weighted toward protein-truncating variants. TRAF7 retained a prior exact-gene local-endpoint count, but its human missense versus loss-of-function endothelial mismatch is stated explicitly and human missense equivalence is not claimed.

2.10 Background clinical signs

Muscle tone, muscle strength and gait were recorded in a designated non-count-eligible layer. These signs are shared endpoints of many unrelated lesions and were not allowed to generate physiological routes. Prevalence, strength and creatine kinase, muscle mass, gait subtype and lesion level were recorded separately where available. No sign was permitted to explain another sign without an independently documented mechanism.

This rule removed a recurrent default inference from gait or hypotonia to reduced skeletal-muscle-pump venous return. Gait may instead reflect cerebellar coordination, sensory afference, basal ganglia sequencing, joint stability, corticospinal control, altered or excessive vestibular signalling, or increased locomotor cost. Low vagal tone or other autonomic abnormalities may coexist with altered posture and exertional tolerance, but neither vagal state nor vestibular direction can be inferred from gait alone. A muscle-to-venous-return route remained admissible only where a named muscle or motor-neuron lesion and direct haemodynamic measurements were available.

2.11 One primary node and one count

Each gene received one controlling primary placement, zero or more measured secondary routes and at most one strict count. Multiple interesting mechanisms could not inflate the numerator. The counted node is named explicitly in every counted row.

2.12 Status classes

For publication presentation, status and strict-count contribution are separated below. The audit workbook preserves the frozen codes Yes / 1, Partial / 0, No / 0, Unresolved / 0 and Cross-ref / 0; the number after the slash denotes contribution to the strict numerator, not a pass/fail score for the gene or the framework.

Status	Operational meaning	Strict-count treatment
Yes: gate cleared	One strict primary gate cleared.	Contributes 1.
Partial: informative hold	Informative biology retained, but the row is structurally non-counting, mismatched in allele, dosage or model, or missing a direct functional, challenge-state or systemic bridge.	Contributes 0; not a negative finding.
No: boundary exclusion	A specific proposed route failed a classifier boundary or serves as a comparator.	Contributes 0; route refused.
Unresolved: mechanism hold	Disease allele or mechanism cannot yet be interpreted safely.	Contributes 0; interpretation held open.
Cross-reference: informational	Bounded informational connection; route generation barred.	Contributes 0; informational only.

2.13 Systems roles and presentation coding

Orthogonally to domain placement, each gene was assessed for three non-exclusive roles: demand or challenge generation; defence-controller or rescue-execution modification; and target-tolerance or recovery modification. Domain records where a gene sits; role records what it does there.

The frozen role fields were structured free text. For presentation only, a deterministic coding was applied after freeze: role fields beginning with "Yes" were classified as assigned or strongly supported; explicit "No" or "weak/no" fields as no direct role; and all qualified or conditional language as candidate or conditional. This display coding has no count effect and does not convert a candidate role into evidence. The number of genes assigned to a role is a frequency of coded observations, not a measure of effect size, causal strength or clinical importance.

2.14 Interpretation of defence responses and challenge-state criteria

An observed physiological response was not assumed pathological. It was classified, where possible, as initiating pathology, successful compensation, failed compensation, chronic over-reliance, failure to stand down or downstream cost.

A gene entered the measured challenge-state Results only where evidence existed in the form baseline -> defined challenge -> altered response magnitude, reserve, recovery or adaptation. Predictions that heat, dehydration, exercise, repeated firing or postprandial stress would expose a phenotype remained in the experimental programme. Normal challenge, recovery and rechallenge across the proposed compartment would weaken the corresponding connection; compensation cannot be invoked indefinitely to reinterpret negative findings.

2.15 Governance, audit and freeze

The analysis was maintained as a versioned working master with a provenance chain and quality-assurance record. Reviewers were ultimately required to record the supplied status, count, domain and master version. Independent reconstruction of a baseline from earlier documents was prohibited after it produced documented errors.

HB-05 Part 1 compared each current row against the final status convention. HB-05 Part 2 compared the baseline stated by each reviewer with the master active at the time of the review, using a fourteen-batch provenance index and five seeded positive controls. Coverage was reported rather than converted into a misleading pass rate.

HB-05 Part 2 coverage class	Rows
Full status comparison verifiable	23
Count-only comparison verifiable	17
Structurally protected because baseline was withheld	16
Coverage-limited or unverifiable	46
Total	102

All five seeded failures were detected. Six mismatch or flag records were identified, all of which had already been re-adjudicated before freeze, and no new numerator movement resulted. Unverifiable rows were not classified as conforming.

The strict numerator was frozen at 42 of 102 under freeze identifier NF-2026-08-11-v46.20 [10-12]. Any later count movement requires a post-freeze amendment recorded in the delta and change ledgers.

2.16 Hierarchical coding of the 56 Partial rows

To describe why the Partial class did not clear the gate, each Partial row was assigned one primary barrier using a frozen, hierarchical presentation rule. First, rows in support, downstream, contextual or unresolved layers were classified as structurally non-counting by placement unless a separate direct

endpoint closed. Among the remaining Partials, rows with an explicit allele, dosage or model mismatch were assigned to that category. The residual rows were classified as missing a direct functional, challenge-state or systemic bridge despite closer disease relevance. The coding is descriptive and has no count effect.

2.17 What the analysis cannot establish

- There is no matched comparison gene set and no formal enrichment statistic.
- The classification was motivated by a physiological framework and was not blind to it.
- The ontology and governance rules evolved during the project, although changes were versioned and retrospectively harmonized.
- Gate types differ in experimental burden, so the count distribution reflects both physiology and measurement opportunity.
- The analysis does not establish that the proposed physiology is present in autistic people.
- The number of genes in a domain, umbrella or systems role does not estimate the magnitude of that domain's physiological effect; a single high-leverage gene may be more consequential than many weakly acting genes.
- No clinical or therapeutic inference follows.

3. Results

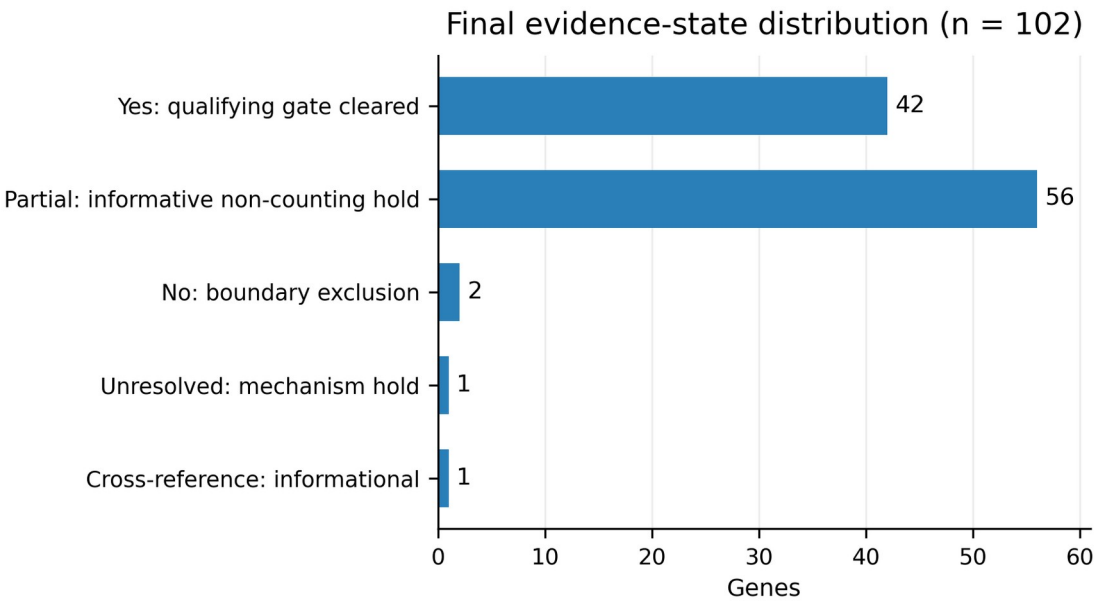
3.1 Final classification

Under the frozen v46.20 rules, 42 genes cleared one strict gate [10-12]. The status distribution is an evidence-state map, not a pass/fail score. The 56 Partial rows retained informative biology but could not legitimately enter the strict numerator: 35 because the primary placement lay in a support or contextual layer that was non-counting by construction, 16 because the evidence carried an allele, dosage or model mismatch, and five because a direct functional, challenge-state or systemic bridge remained unmeasured. Two rows were explicit boundary exclusions, one was unresolved and one was a bounded cross-reference. A No status records refusal of a specific proposed route at the classifier boundary; it is not a gene-level disproof of the hypothesis.

Table 2. Final evidence-state distribution in the frozen v46.20 map [10-12]. Status and strict-count contribution are separated so that the audit code is not read as a pass/fail score.

Status	Genes	Interpretation
Yes: gate cleared	42	Qualifying gate cleared; strict-count contribution = 1.

Status	Genes	Interpretation
Partial: informative hold	56	Informative non-counting holds; strict-count contribution = 0. Support/context placement by construction (35); allele, dosage or model mismatch (16); or missing functional, challenge-state or systemic bridge (5). These are not negative findings.
No: boundary exclusion	2	LRRC4C and SMARCC2: explicit boundary exclusions of proposed routes; strict-count contribution = 0.
Unresolved: mechanism hold	1	ASXL3: held for allele-mechanism uncertainty; strict-count contribution = 0.
Cross-reference: informational	1	SATB1: bounded chloride-maturation cross-reference; strict-count contribution = 0.



Statuses describe evidence eligibility under the frozen rubric, not pass/fail outcomes.

Figure 1. Final evidence-state distribution [10]. These statuses describe evidence eligibility under the frozen rubric, not pass/fail outcomes: Partial denotes an informative non-counting hold, and No denotes a boundary exclusion of a proposed route.

3.2 Physiological occupancy of the complete gene set

The seven publication umbrellas summarize the primary placement of all 102 genes without exposing the main text to 33 technical bins [10]. The largest occupancy was the deliberately non-counting umbrella of immune, epithelial, recovery and contextual layers (33 genes; strict-count conversion not applicable by design). Among physiological umbrellas, chloride, inhibitory maturation and ionic excitability contained 20 genes, placental/vascular/cardiac/autonomic/mechanical support 13, secretory/circadian/metabolic response 15, astrocytic/osmotic/CSF/energetic recovery 9, renal/mineral/endocrine control 7, and structural/membrane/extracellular containment 5.

Figure 2 compares primary occupancy with strict counts. These bar lengths do not measure the strength, rarity or physiological leverage of a category; a sparsely populated category may contain a high-impact controller.

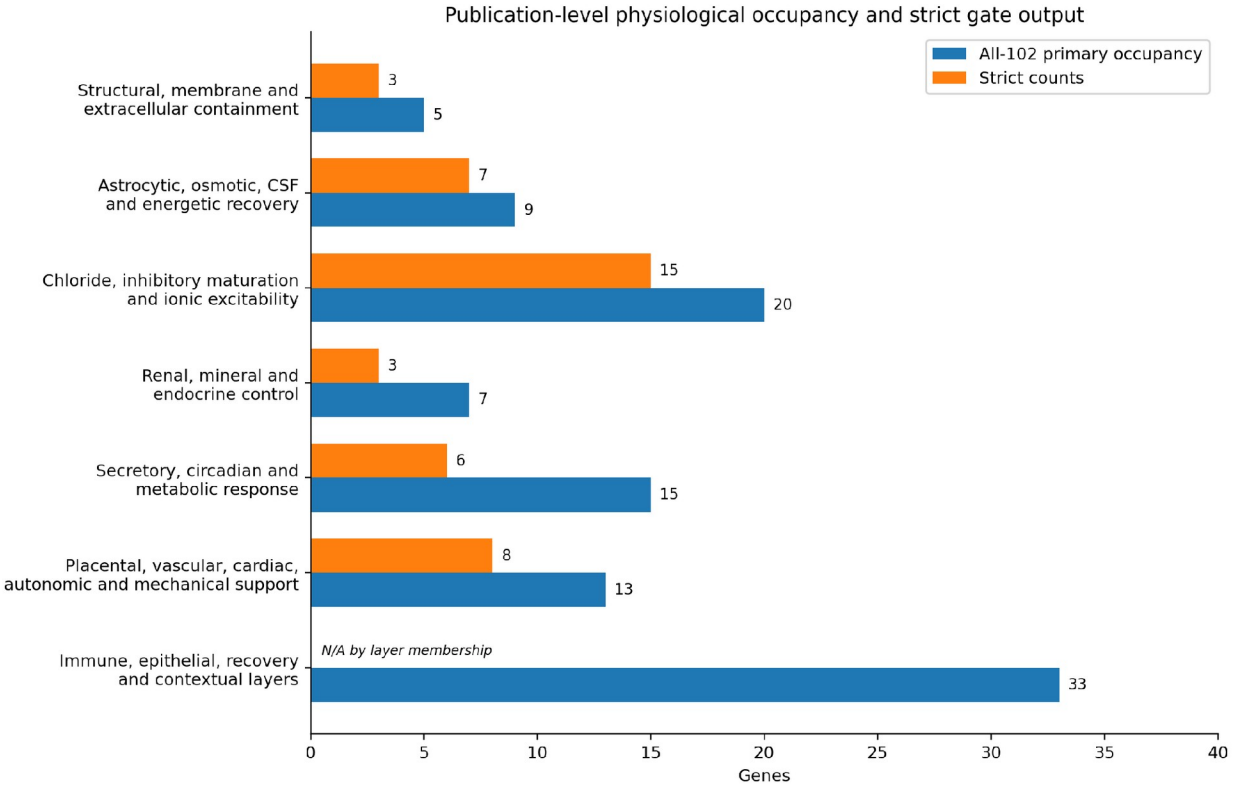


Figure 2. Primary occupancy and strict gate output across seven publication umbrellas [10]. The umbrellas are presentation categories within one coupled architecture, not a core-versus-support hierarchy. The final support/context umbrella is structurally non-counting, so a conversion percentage is not applicable. Occupancy is not an effect-size estimate.

3.3 Strict gate output by qualifying endpoint family and relation to the hypothesis

The 42 counted genes do not represent a single physiological claim. They clear heterogeneous local or domain-specific gates. Reporting the numerator by endpoint family prevents direct ion-current evidence, mineral-endocrine set-point physiology and local vascular-development endpoints from being collapsed into one category. The ten families provide the finer breakdown needed for interpretation without reopening the frozen technical ontology. Figure 2 and Table 3 therefore answer different questions: the publication umbrellas merge related physiological territory for occupancy display, whereas the gate-family table keeps qualifying endpoint types separate.

Table 3. Strict numerator by qualifying endpoint family [10,11]. AP2S1 is a mineral-endocrine count, not direct sodium, chloride or water-handling evidence. Complete gene-level primary-source locks for all 42 counted endpoints are provided in Supplementary Data S2 [11].

Gate family	n	Counted genes	Qualifying endpoint
Direct ion transport and excitability	12	CACNA1E, CELF4, DPYSL2, GRIA2, GRIN2B, HECTD4, KCNMA1, KCNQ3, MYT1L, SCN1A, SCN2A, SYNGAP1	Current, gating, receptor function, channel trafficking or state-dependent excitatory gain.
Chloride persistence and inhibitory signal handling	3	GABRB2, GABRB3, SLC6A1	GABA-A chloride-current readers/effectors or ion-gradient-dependent GABA clearance.
Structural and membrane integrity	3	ANK2, CHD8, SHANK3	Axonal/membrane scaffold, myelin/conduction or receptor-microdomain integrity.
Energetic and mitochondrial recovery	2	IRF2BPL, MBD5	Direct mitochondrial respiratory or high-cost energetic-recovery phenotype.
Neural ECS, astrocyte and CSF interfaces	5	GFAP, PAX5, PTEN, PTK7, RFX3	Astrocyte/ECS mechanics, ventricular/CSF interface or flow hardware.
Renal, mineral and neuropeptide set-point	3	NR3C2, AP2S1, FOXP1	Renal mineralocorticoid execution, CaSR/mineral set point, or AVP/OXT magnocellular-cell development.
Secretory and metabolic response	6	STXBP1, ARID1B, BCL11A, DYRK1A, GIGYF1, TCF7L2	Rapid deployment, beta-cell/insulin/IGF or somatotropic response reserve.
Cardiac/volume effector reserve	1	NUP155	Atrial electrical and beta-adrenergic cardiac effector reserve.
Local vascular and perfusion architecture	4	CTNNB1, TEK, TRAF7, VEZF1	Retinal/venous/lymphatic architecture, endothelial shear response or local outflow reserve.
Brainstem, autonomic and respiratory control	3	DSCAM, PPP1R9B, TCF4	Respiratory rhythm/chemosensory gain or measured central autonomic/baroreflex physiology.

Within the Sodium-Struggle Hypothesis, these endpoint families can converge without all performing the same function. Some increase ionic or metabolic load; some alter renal, endocrine, cardiac, autonomic or secretory defence and execution; some affect extracellular, vascular or lymphatic containment, central filling and perfusion; and others determine how well neural and peripheral tissues recover. The hypothesized route to autism is developmental and systems-level: repeated or poorly resolved instability across these upstream functions could alter cerebral delivery, extracellular and ionic stability, inhibitory maturation and recovery during sensitive windows. This classification identifies candidate routes into that sequence; it does not demonstrate that the sequence occurs in any individual.

3.4 Demand and controller roles against a broad recovery background

The most discriminating systems-role assignments were demand or challenge generation (29 genes) and defence-controller or rescue-execution modification (34 genes). These roles identify relatively specific positions in the proposed architecture. By contrast, target-tolerance or recovery was strongly assigned

to 78 genes. A role applying to 76% of the set has limited discriminatory power, so recovery is treated here as a broad background condition rather than as an independent convergence result. Candidate or conditional roles remain in the completed Systems Role Matrix, but they do not affect the strict numerator.

To test whether the recovery coding was arbitrary, the 24 genes outside the strongly assigned recovery set were examined separately. They were not a random residue. They were concentrated in recognizable groups: controller or set-point genes whose principal measured effect was command rather than target recovery (for example AP2S1, CREBBP, DEAF1, DSCAM, GNAI1, PPP1R9B, RORB and TCF4); vascular or demand-generating architecture genes without a direct target-recovery endpoint (including SKI, TEK, TRAF7 and VEZF1); and unresolved or contextual genes for which disease-matched recovery evidence was insufficient (including ANKRD11, ASXL3, KMT2C, NSD1, PHF21A, SMARCC2 and TAOK1). Load-focused genes such as HECTD4, MYT1L and SLC6A1 also remained outside the strongly assigned recovery set because recovery itself had not been measured. This coherence suggests that the coding is not arbitrary, but the high prevalence still makes recovery a poor headline discriminator. The paper therefore foregrounds demand and controller assignments; recovery is retained as contextual structure, and the full role intersections remain available in the supplementary Systems Role Matrix.

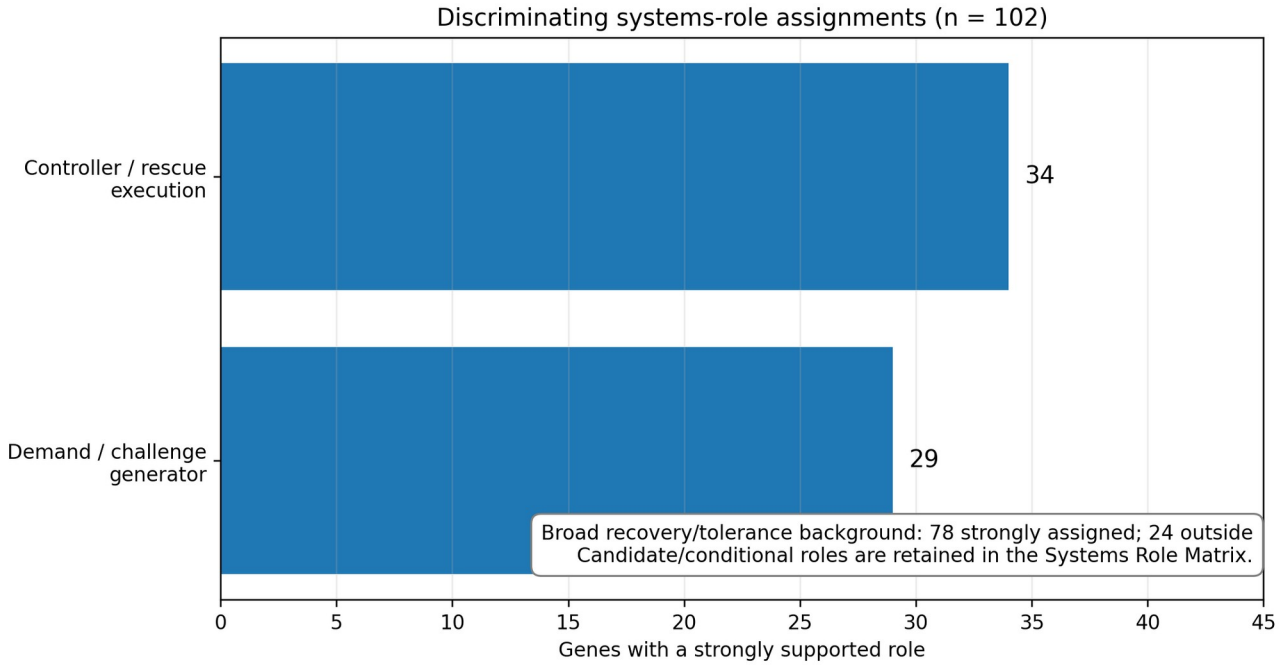


Figure 3. Strongly supported demand and controller roles are foregrounded because they are more discriminating [10]. Recovery or target-tolerance was strongly assigned to 78 genes and is treated as a broad background condition; the 24 genes outside that role were examined separately for coherence. Role frequencies do not estimate effect size and do not affect the strict numerator.

3.5 Sparse primary placement in canonical renal/RAAS/AVP and transporter-level sodium handling

At the level of primary placement, the narrow canonical renal/RAAS/AVP and transporter-level sodium-handling branch was represented but sparse. Renal Salt and Water Handling contained one gene (NR3C2); Direct Vasopressin and Oxytocin Biology one (FOXP1); and Controller Receptor, Coupling and Signalling one, not counted. Volume, Cardiac Function and Vascular Containment contained two genes, one counted (NUP155), but this domain belongs to the circulatory arm of the broader architecture

rather than lying outside sodium-volume defence. By comparison, 14 genes occupied Direct Ion Transport and Excitability Load and 12 Cellular Recovery Infrastructure.

This is sparse primary placement in the narrow canonical branch, not sparse representation of the whole sodium-volume-perfusion defence architecture. Vascular and lymphatic containment, cardiac output and reserve, autonomic and respiratory control, secretory execution, ionic workload, energetic reserve and cellular recovery occupy other positions within the same coupled system or modify its load and recoverability. The one-primary-node rule also means that measured secondary routes into machinery domains exist for genes whose primary placement lies elsewhere. No enrichment claim is made.

3.6 Measured challenge-state and recovery phenotypes

The evidence-register IDs in Tables 4 and 5 identify versioned workbook finding records. They are audit identifiers, not bibliographic citations. Primary citations are provided in Tables 4-6, and complete gene-level source locks for the counted endpoints are preserved in Supplementary Data S2 [11].

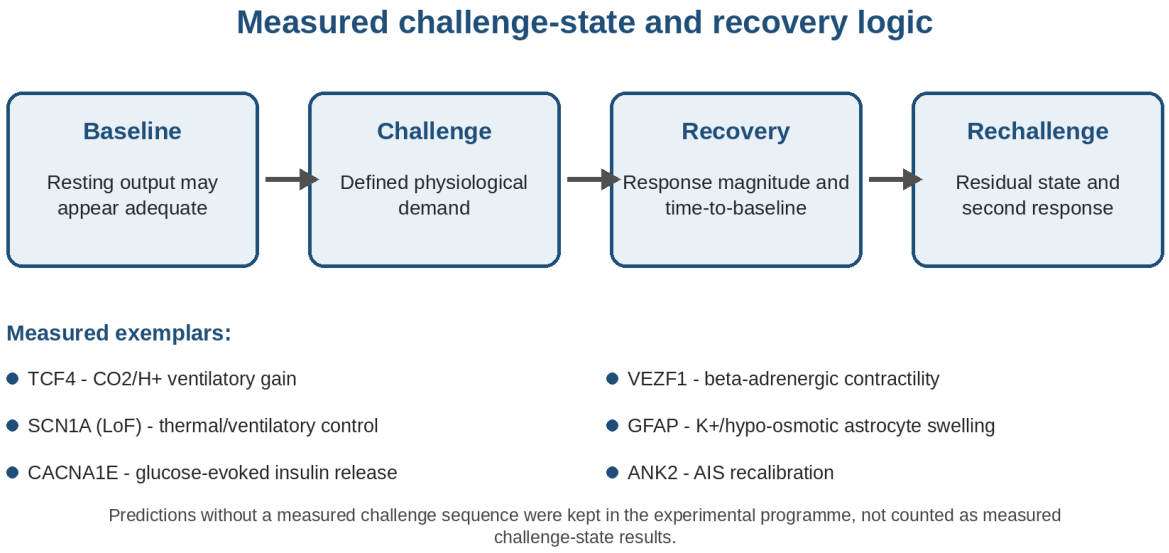


Figure 4. Challenge-state logic used in the classification. Measured exemplars are shown at readable print scale. Predictions without a measured challenge sequence entered the experimental programme rather than the measured challenge-state Results.

Table 4. Measured challenge-state exemplars. The table separates baseline, challenge, response and model boundary. Primary citations support the measured phenotype; evidence-register IDs preserve the internal audit trail.

Gene	Baseline	Defined challenge	Measured deficit	Boundary	Evidence-register ID
TCF4	Comparatively preserved baseline averages and hypoxic responsiveness	Hypercapnic CO2/H+ challenge	Markedly impaired ventilatory reserve, unstable breathing and prolonged post-sigh apnoea [13]	Heterozygous disease-relevant model; no systemic volume bridge.	SG2-TCF4-01 to -07
SCN1A, LoF branch	Near-normal resting core temperature and respiratory output in relevant models	Sustained heat/fever and ventilatory challenge	Impaired temperature-sleep coupling and abnormal ventilatory control [14,15]	Branch-specific LoF physiology; not automatically ASD-missense physiology.	SG6-SCN1A-01 to -09

Gene	Baseline	Defined challenge	Measured deficit	Boundary	Evidence-register ID
CACNA1E	Normal fasting glucose and insulin in Cacna1e-null mice	Glucose challenge	Reduced glucose-evoked second-phase insulin release and granule recruitment [16]	Homozygous-null physiology; disease-allele whole-body reserve untested.	SG14-CACNA1E-01 to -07
VEZF1	Baseline anatomy or contractility can appear partly preserved	Beta-adrenergic cardiac stimulation	Limited normal contractility increase under beta-adrenergic stimulation [17]	Exact-gene model; systemic filling and human disease-dose translation unmeasured.	SG1-VEZF1-01 to -06
GFAP	Baseline extracellular-space parameters can appear similar	High extracellular K+, hypo-osmotic stress or cortical spreading depression	Altered astrocyte swelling and attenuated challenge-induced extracellular-space narrowing [18]	Null physiology; toxic Alexander-disease variants may differ.	SG11-GFAP-01 to -10
ANK2	Resting AIS structure and network activity are abnormal but stable	Sustained activity increase or decrease	Impaired homeostatic AIS length plasticity [19]	Disease-class-matched heterozygous LoF neural model; systemic reset not inferred.	SG14-ANK2-01 to -08

3.7 Representative gene-level exemplars

Fourteen exemplars were selected to span endpoint families rather than to favour the largest domain. Each is summarized in the same four-part structure: measured anchor, conservative systems meaning, missing joint and decisive experiment.

Table 5. Representative gene exemplars spanning the major qualifying endpoint families. Primary citations support the measured anchors; evidence-register IDs preserve the internal audit trail.

Gene / evidence-register ID	Measured anchor	Conservative systems meaning	Missing joint and decisive experiment
CELF4 SG13-CELF4-01 to -06	Celf4 deficiency increases cortical AIS Nav1.6 and persistent sodium current [20].	Compartment-specific excitability calibration with a direct cortical ionic-load endpoint.	Measure Na/K-pump current, ATP cost and chloride recovery during repeated activity.
SCN2A SG6-SCN2A-01 to -08	ASD-associated variants have directly measured loss-of-function current; severe NaV1.2 loss can produce paradoxical burst-frequency hyperexcitability [21,22].	Clean ascertainment-to-current match with branch-specific dendritic reserve and network load.	Measure cumulative ionic traffic, potassium recruitment and metabolic recovery in variant-resolved neurons.
GRIA2 SG11-GRIA2-01 to -07	Pathogenic variants alter AMPAR current amplitude, rectification and permeability; a gain-of-function variant slows desensitisation and prolongs responses [23,24].	Current duration and integrated charge are distinct load variables.	Measure Na/Ca, ATP and Na/K-ATPase recovery during repeated glutamate trains.
GABRB3 SG11-GABRB3-01 to -08	Pathogenic heterozygous variants produce allele-specific gain- or loss-of-function effects on GABA-A receptor current, sensitivity, gating and desensitisation [25].	Allele-specific chloride-current reader with direct receptor-function evidence.	Measure chloride load, E_GABA, integrated current and recovery during repeated activation in disease-variant neurons.
PPP1R9B SG8-PPP1R9B-01 to -07	Spinophilin loss produces high blood pressure and heart rate, impaired baroreflex buffering, increased central sympathetic outflow and reduced parasympathetic influence [26].	Direct machinery-side autonomic/controller phenotype.	Repeat at disease dosage with telemetry, baroreflex unloading and receptor/pump surface-residency assays.

Gene / evidence-register ID	Measured anchor	Conservative systems meaning	Missing joint and decisive experiment
TCF4 SG2-TCF4-01 to -07	Heterozygous models show abnormal parafacial/RTN CO2/H+ challenge gain and respiratory-pattern instability [13].	Resting averages can conceal compressed controller dynamic range.	Add simultaneous ventilation, BP, stroke volume and endocrine outputs during hypercapnia.
STXBP1 B17-SOL2-STXBP1-01 to -04	Munc18-1 is required for neuronal dense-core-vesicle exocytosis; heterozygous reduction preferentially impairs the early high-demand phase [27].	Neuronal secretory-deployment reserve rather than peptide-production deficit.	Test disease variants in neuronal and neuroendocrine pulse-washout-rechallenge assays.
GFAP SG11-GFAP-01 to -10	GFAP loss alters astrocyte swelling and extracellular-space responses to K+, hypo-osmotic and spreading-depression challenges; GFAP also regulates astrocyte mitochondrial fission [18,28].	Local CNS microenvironment-reset and recovery-bandwidth gene.	Compare ASD variants, null and toxic-mutant controls under ionic, osmotic and glutamate challenge.
CHD8 SG13-CHD8-01 to -07	Disease-relevant Chd8 haploinsufficiency produces oligodendrocyte/myelin deficits and increased intestinal amino-acid transporter expression [29,30].	Myelin/conduction reserve with a separate gut-interface and substrate route.	Measure conduction Na/ATP recovery and independently test gut/mesenteric physiology.
CTNNB1 SG13-CTNNB1-01 to -08	Human heterozygous CTNNB1 variants cause retinal vascular-development failure; endothelial Ctnnb1 deletion disrupts adult blood-brain barrier integrity [31,32].	Local vascular-delivery and barrier reserve.	Measure disease-variant endothelial barrier threshold, regional flow and recovery under challenge.
TEK SG2-TEK-01 to -08	Human activating TIE2/TEK variants cause inherited and cutaneomucosal venous malformations [33,34].	Allele-selective local venous architecture.	Measure disease-branch venous compliance, regional filling, barrier challenge and systemic haemodynamics.
VEZF1 SG1-VEZF1-01 to -06	Vezf1 loss alters cardiac structure and beta-adrenergic contractile reserve; reduced dosage also disrupts vascular development [17,35].	Local lymphatic architecture plus cardiac rescue-execution reserve.	Measure lymph propulsion, plasma volume, filling and cardiac output during graded challenge.
ARID1B SG14-ARID1B-01 to -07	Low IGF1 fails to recruit the expected GH rise despite preserved GHRH-triggered GH release; GH rescue is effective [36].	Somatotropic feedback-controller recruitment failure.	Test low-salt/dehydration challenge with GH rescue, urine sodium, ECW, RAAS, AVP and haemodynamics.
AP2S1 SG14-AP2S1-01 to -07	Arg15 AP2S1 variants alter CaSR trafficking and mineral-endocrine set point; generic dosage loss does not reproduce the same phenotype [37,38].	Allele-selective receptor-trafficking set-point control.	Test NDD alleles across a receptor cargo matrix and controlled sodium/volume challenge.

3.8 Architecture of refusal

Four genes are presented because each failed or triggered the gate for an independent reason. The purpose is to describe the classifier rather than defend it: preserved phenomenon, failed criterion, and measurement that would change the answer.

Table 6. Architecture of refusal and revocation. ASXL3 entries include two 2026 preprints and are treated as provisional [41,42].

Gene	Firewall	Preserved finding	Failed criterion	Measurement that changes the answer	Treatment
LRRC4C	Compartment	Stress-responsive circuit and widespread neuronal-activity abnormalities [39].	Neural challenge responses are not arterial baroreflex or vascular mechanosensing.	Demonstrate LRRC4C-dependent arterial baroreceptor or vascular mechanosensor execution during haemodynamic challenge.	No: boundary exclusion
SMARCC2	Evidence type	BAF170/SMARCC2 loss alters cardiac-lineage chromatin accessibility and delays the onset of beating in differentiating cells [40].	Chromatin accessibility and delayed beating are not measured ion current or flux at disease-relevant dosage.	Direct patch clamp or flux measurement under challenge in disease-relevant models.	No: boundary exclusion
ASXL3	Allele mechanism	Preprint evidence links Asxl3 loss to DIO3/local-T3 dysregulation in parvalbumin interneurons [41].	A second preprint reports that some pathogenic truncations may escape nonsense-mediated decay and produce persistent truncated protein; null and truncation models cannot be safely pooled [42].	Isoform- and product-specific mapping of patient truncations with local-T3 and physiological outputs.	Unresolved: mechanism hold
DYNC1H1	Ascertainment revocation	Mitochondrial morphology/function and stress-responsive organelle positioning are impaired in DYNC1H1 missense models [43].	Qualifying physiology was missense-model based while ASD ascertainment was PTV/motor-domain weighted.	PTV-specific mitochondrial or transport phenotype, or exact ASD-variant match.	Count revoked: 1 -> 0

3.9 The 56 Partial holds

The 56 Partial rows should not be interpreted as 56 negative tests of the framework. A Partial preserved informative biology but marked that the row could not legitimately enter the strict numerator under the frozen rules. For 35 rows, the primary placement itself lay in a support, downstream, contextual or unresolved layer that was non-counting by construction unless a separate direct endpoint closed; these rows were not tested against a count-eligible primary gate and found negative. Of the 21 Partials in count-eligible or exploratory domains, 16 were blocked by allele, dosage or disease-model mismatch, and five had closer disease relevance but lacked a direct functional, challenge-state or systemic bridge.

Table 7. Hierarchical primary-barrier coding of the 56 Partial rows. Categories are mutually exclusive for presentation, although individual genes may have several biological limitations.

Primary barrier	Genes	Examples or scope	Interpretation
Support/context placement by design	35	Domains 25, 26, 27, 30, 31, 32 and 33.	The primary row preserves illness amplification, circuit timing, cellular recovery or contextual development without treating those layers as direct physiological gate output.

Primary barrier	Genes	Examples or scope	Interpretation
Allele, dosage or model mismatch in a count-eligible/exploratory domain	16	Examples: DYNC1H1, SKI, TLK2, TRIM23, CREBBP, DEAF1.	The measured phenotype exists, but the evidence-carrying allele, dosage, species, transformed-cell or tissue-specific model does not match the ASD disease branch closely enough.
Closer disease relevance but missing direct functional or systemic bridge	5	FOXP2, NCOA1, NRXN1, POGZ, RAI1.	The gene-specific route is informative, but the qualifying functional output, challenge response or local-to-systemic joint has not been measured.

This distribution changes the interpretation of a zero. In the status code, zero denotes strict-count contribution, not biological failure. Most zeros do not mean that no relevant biology was found. They mean that the field has measured a different dosage, a different compartment, anatomy without function, molecular mechanism without flux, or a plausible route without the challenge and systems measurements required by the gate.

3.10 Prioritised experimental programme

Table 8. Priority experimental programme organized by physiological problem rather than gene alphabet.

Experimental class	Core measurements	Representative genes
Variant-matched ion and recovery assays	Direct current plus intracellular Na+, Ca2+ and Cl-, ATP/ADP, pump current, E_GABA, cell volume and time-to-baseline.	CELF4, GRIA2, SCN2A, SYNGAP1, CHD2, CACNA1E
Secretory pulse-washout-rechallenge	Separate production, storage, release, surface delivery, clearance and receptor response.	STXBP1, FOXP1, GIGYF1, TCF7L2, BCL11A, CACNA1E
Controlled volume, heat and dehydration challenge	Plasma volume, stroke volume, regional venous volume, urine sodium/osmolality, RAAS, copeptin/AVP and recovery.	NR3C2, ARID1B, FOXP1, TEK, VEZF1, PPP1R9B
Vascular and lymphatic reserve	Shear response, barrier threshold, compliance, regional flow, lymph propulsion and systemic filling.	CTNNB1, TRAF7, TEK, VEZF1, FOXP2
Gut/ENS and splanchnic direction	Motility, epithelial loss, stool water/sodium, meal redistribution, mesenteric flow and venous capacitance.	CHD8, FOXP1, TCF20, TM9SF4, DEAF1
Muscle force versus venous return	Strength, CK, muscle mass, EMG, repeated force, venous emptying/refill, preload and stroke volume.	ADNP, PHF2, SHANK3, KMT5B, ANK2
Allele-mechanism and compartment resolution	Patient-like knock-ins, exact mutant-product mapping, tissue-specific rescue and same-model comparison.	ASXL3, DYNC1H1, TRAF7, AP2S1, CORO1A, CREBBP
Uniform-depth physiological screening	Identical clinical chemistry, cardiovascular and metabolic parameters across all available gene models.	All 102, using standardized consortium data where available

4. Discussion

4.1 Distributed upstream convergence across brain and body

Taken together, the fixed Satterstrom-102 set maps onto a distributed upstream architecture relevant to the coupled sodium-volume-perfusion defence problem posed by the Sodium-Struggle Hypothesis. The hypothesis is not a renal-first sequence in which a sodium disturbance occurs and vascular, cardiac or

autonomic systems merely support it. Rather, the defended problem spans extracellular tonicity, effective central filling, vascular tone, capacitance and containment, regional distribution and tissue perfusion, with renal, endocrine, cardiac, vascular and autonomic sensing and controller recruitment and with cellular execution and recovery. The measured gene-specific phenotypes occupy different entry and control points: osmotic or ionic demand, filling and forward flow, containment and regional allocation, cardiac or respiratory reserve, controller execution, energetic cost and target-tissue recovery. Many of these functions are expressed in peripheral organs or whole-body control systems, not only in the brain.

In the causal ordering of the hypothesis, most of these mapped functions sit upstream of its proposed downstream neural and developmental consequences. Depending on the measured phenotype, a gene may generate challenge, alter sensing or controller execution, constrain structural or energetic reserve, change regional delivery, or reduce recovery. Not every gene in one of these broader domains is itself a direct sodium-volume controller. The numerical distribution cannot determine the magnitude of those effects. One high-leverage controller may matter more than many weak modifiers, and category size is influenced by measurement opportunity. The defensible conclusion is therefore about the diversity and position of the routes, not their relative causal weight.

4.2 Measurement opportunity and evidence-type bias

This is the most important limitation of the distribution. Gate families differ in experimental burden. A direct ion-current phenotype can often be established in one carefully designed voltage-clamp experiment. Demonstrating abnormal effective central filling, renal challenge response, splanchnic mobilisation, cardiac reserve, circadian endocrine scheduling or defence termination may require a disease-matched model, controlled physiological challenge, simultaneous measurement across several organ systems and recovery or rechallenge sampling.

The strict count distribution therefore reflects at least two things: underlying biology and the availability of sufficiently direct gene-specific measurements. A support or contextual layer with no counts is not a domain with a 0% biological pass rate. It is a layer whose membership is non-counting by design or whose qualifying experiment has rarely been performed.

4.3 Literature-depth and model bias

The literature is differentially investigated. Highly studied neuronal genes may have extensive synaptic or developmental data but little renal, vascular, endocrine or autonomic phenotyping. Less familiar genes may be studied only in cancer cells, null embryos, screening platforms or severe conditional models. This creates two opposing errors: over-interpreting a convenient model, and interpreting absence from the literature as absence of physiology. The present classifier reduces but cannot eliminate that bias.

4.4 Relation to existing ASD convergence work

The present analysis addresses a different level from molecular and developmental convergence studies. It does not compete with evidence for shared cortical progenitor, transcriptional or synaptic programmes. Instead, it places the peripheral and whole-organism phenotypes associated with the same gene set into a common evidence structure. Whether those phenotypes are incidental comorbidity, consequences of severe neurodevelopmental disease or part of the same causal architecture remains open.

4.5 What the 42 represents

The number 42 is a frozen rubric-based count, one primary endpoint per gene, drawn from heterogeneous gate families. It is not an estimate of how many autism-risk genes directly cause systemic sodium or volume abnormality, and it does not rank categories by physiological strength. The numerator-by-gate table is therefore essential: direct current, mineral-endocrine set point, local vascular development, central fluid-space hardware and respiratory-controller gain are not interchangeable observations.

AP2S1 illustrates the point. Its count is justified under the mineral-endocrine gate through allele-selective AP2S1-CaSR trafficking and mineral set-point physiology [37,38]. It should not be described as direct AP2S1 sodium, chloride or water handling.

4.6 Why the Partial class is scientifically useful

The Partial class converts uncertainty into an experimental programme. A gene can have strong and relevant physiology without clearing the strict gate. The Yes/Partial boundary is an evidence-eligibility distinction under the rubric, not a biological partition between genes that participate and genes that do not. Each Partial row must identify the barrier to counting and a test capable of proving or killing the proposed connection. The hierarchical analysis shows that most Partials are either placed in non-counting support/context layers or blocked by a specific allele/model problem. Only five count-eligible or exploratory Partials had comparatively close disease relevance but still lacked the direct functional or systemic bridge.

The 56 Partials therefore specify what the field has not yet established at the qualifying level: disease-allele physiology, heterozygous rather than null response, function rather than anatomy, flux rather than expression, challenge rather than baseline, or systemic consequence rather than local plausibility.

4.7 Challenge-state physiology

Measured challenge-state phenotypes were one of the clearest cross-gene patterns. TCF4, the SCN1A loss-of-function branch, CACNA1E null models, VEZF1, GFAP null models and ANK2 showed different versions of the same logic: resting output can be comparatively adequate while response gain, high-load performance, recovery or structural recalibration is impaired [13-19].

This observation justifies baseline-challenge-recovery-rechallenge designs in neurodevelopmental physiology. It does not justify reinterpreting every normal baseline as masked disease. The design remains falsifiable: normal response magnitude, recovery and rechallenge in the relevant compartment weaken the proposed connection.

4.8 Relation to the Sodium-Struggle Hypothesis

The observed distribution is compatible with, but does not exclusively support, the Sodium-Struggle Hypothesis. The core hypothesis was developed beginning in 2015, independently of and before the Satterstrom-102 study, so it was not constructed from these genes. Genetic variation is also not required by the hypothesis. Within the fixed 102, primary placement in the narrow canonical renal/RAAS/AVP and transporter-level sodium-handling branch was sparse rather than absent. This should not be read as sparse representation of the broader sodium-volume-perfusion defence architecture. Other primary and secondary routes occupied vascular and lymphatic containment, cardiac output and reserve, autonomic and respiratory control, endocrine and secretory execution, regional delivery, ionic and energetic load, and cellular recovery. These are components or modifiers of the same coupled architecture, although not every gene in a broader domain is itself a direct controller. One interpretation is that autism-associated variants often modify challenge generation, controller

execution, containment, distribution, reserve or recovery, making a Sodium-Struggle state more likely or harder to reset without constituting the initiating event.

Alternative explanations must be given equal weight. The distribution may reflect what neuroscience and cell biology have measured; the classifier may recognize direct current and local structural endpoints more readily than multi-organ reserve; the seven umbrellas may group mechanisms with unequal effect sizes; and another neurodevelopmental gene set may produce a similar systems pattern. The present study therefore supplies a bounded comparison with the hypothesis and a set of discriminating experiments, not confirmation of an exclusive causal theory.

4.9 Mutation consequence and mutation origin: scope and timing firewall

Mutation consequence and mutation origin are distinct questions. The present phenotype-first analysis evaluates physiological consequences after disruption of a fixed set of autism-risk genes; it does not determine why an underlying variant arose, in which cell it arose, whether it arose in a parent or child, when it arose, whether a physiological exposure altered its formation, or whether selection rather than mutation formation influenced its observation. De novo status does not establish developmental timing or cell of origin. Most constitutional human de novo variation is established before the child's later postnatal physiology. A multigenerational study in another cohort found a strong paternal bias among germline de novo mutations and also identified a minority of postzygotic events [44]. That cohort-level distribution is not assigned to the Satterstrom events, whose parental origin and developmental timing have not been resolved in this project and are not inferred event by event.

A child's postnatal Sodium-Struggle physiology cannot generate an inherited allele, a variant already present in parental germ cells, or a constitutional variant established before the relevant physiological window. A genomic-vulnerability observation is informative about mutation origin only when the measured cell type and developmental window are compatible with the time and place in which the mutation could have formed; fetal-neural observations therefore cannot automatically explain variants that arose in parental germ cells. A mutation could in principle be downstream of one earlier biological process and subsequently become upstream of later dysfunction, but this study does not determine whether that sequence applies to any Satterstrom variant. Parental-germline effects, early postzygotic instability, tissue-restricted mosaicism, common-upstream stress, state-dependent repair, broad mutagenesis followed by selection, and lineage selection without increased mutation formation are distinct hypotheses requiring separate timing- and cell-type-matched analyses. No individual Satterstrom variant is classified here as having been caused by Sodium-Struggle physiology, and no physiological zero, Partial classification or boundary exclusion is interpreted as a genomic scar. Restricting the present analysis to mutation consequence is a strength rather than a limitation: it allows the physiological classification to stand independently of any future account of mutation origin.

4.10 The value of refusal and revocation

The architecture of refusal demonstrates that compelling biology was not automatically counted. LRRC4C failed on compartment, SMARCC2 on evidence type, ASXL3 on unresolved allele mechanism and DYNC1H1 on ascertainment mismatch. DYNC1H1 is especially informative because the gate removed a previously granted count. A classifier that can revoke a result under its own later audit is less compatible with one-directional confirmation than a system that only adds genes.

4.11 What would change the status distribution and numerator

The frozen 42/102 numerator is a versioned decision under stated rules, not an assertion that the number can never change. An upward change would require a non-counted gene to close a qualifying

gate with disease-relevant, gene-specific evidence: for example, a Partial could acquire the missing direct functional, challenge-state or systemic endpoint; ASXL3 could leave the Unresolved class after its allele mechanism is resolved; or SATB1 could move beyond Cross-ref if disease-dose chloride physiology were measured.

A downward change would occur if a counted endpoint failed replication, a model-integrity problem invalidated the qualifying evidence, a disease-allele or direction mismatch proved disqualifying under a symmetrically applied rule, or the endpoint were shown not to belong to the counted gene. DYNC1H1 demonstrates that revocation is possible. By contrast, adding a creative secondary route, moving a gene between publication umbrellas, or finding more biology in a structurally non-counting layer would not change the numerator. Because v46.20 is the freeze point, every later change must be recorded as a post-freeze amendment in the delta and change ledgers rather than as a silent edit.

4.12 Limitations

- There was no matched comparison gene set and no formal enrichment analysis.
- The analysis was motivated by a physiological framework and was not blind to it.
- The technical ontology and governance rules evolved during the project, although changes were versioned and retrospectively harmonized.
- Gate families differ in measurement burden, producing evidence-opportunity bias.
- Literature depth, species, dosage, developmental window and compartment were uneven across genes.
- The number of genes assigned to a domain, umbrella or systems role is an occupancy measure and does not estimate effect magnitude or causal importance.
- Some counted rows retain visible allele or model mismatches under prior adjudicated rules, including TRAF7.
- The one-primary-node rule simplifies genes with multiple measured physiological routes.
- HB-05 Part 2 had incomplete historical coverage: 46 of 102 rows were coverage-limited or unverifiable.
- The role coding is a post-freeze presentation normalization of free-text fields rather than an independent experimental endpoint.
- No new physiological measurements were collected, and the study does not demonstrate a shared sodium-volume state in autistic people.
- No clinical or therapeutic inference is supported.

4.13 Falsification and next steps

The highest-value next step is direct, variant-matched challenge physiology rather than another annotation pass. Experiments should measure ionic traffic and recovery in the same cells; separate hormone production, storage, secretion, surface delivery, clearance and receptor response; quantify effective volume, regional pooling and cardiac output during controlled challenges; and compare disease-relevant alleles rather than importing null, family or paralogue physiology.

5. Conclusion

A gated phenotype-first audit of the fixed Satterstrom-102 gene set, conducted in relation to the independently developed Sodium-Struggle Hypothesis, produced a version-controlled systems-physiology map with a frozen strict numerator of 42/102. The result is not a claim that 42 genes directly cause low sodium or low blood volume. It is evidence that heterogeneous autism-risk genes can occupy upstream positions in a shared physiological architecture involving ionic and metabolic demand; renal, endocrine, cardiac, vascular and autonomic sensing and control; effective filling, regional distribution and tissue delivery; structural and energetic reserve; and target-tissue recovery.

Most non-counted genes retained useful biology and explicit experiments; non-counted is not synonymous with negative. The value of the classifier is therefore not only the numerator. It is the disciplined separation of what is measured, what is inferred, what is refused, how the gene could relate to the Sodium-Struggle architecture, and what observation would change the answer.

Declarations

Ethics approval

This study analysed published human and experimental findings and did not involve new human participants, identifiable private data or new animal experiments.

Funding

This work was self-funded by the author.

Competing interests

The author is an independent, unremunerated researcher with no institutional affiliation, no external funding and no financial interest in the outcome. The author declares one non-financial interest: the physiological framework that motivated this classification is the author's own, and the analysis was undertaken in part to determine whether the fixed gene set was consistent with it. The classification was therefore not blind to the framework. The methodological safeguards used to constrain this interest are described in Methods Sections 2.3, 2.8, 2.9 and 2.15 and are restated in the limitations.

Author contributions

Kathleen R. Bell conceived the Sodium-Struggle Hypothesis, formulated the comparison with the fixed Satterstrom-102 set, guided the selection and refinement of the physiological bins, directed the evidence-mapping and audit process, reviewed and approved the final scientific interpretations, and wrote and revised the manuscript. Bell is the sole author and takes responsibility for the work in its entirety.

AI-assisted research and drafting disclosure

Multiple large-language-model systems were used as research and drafting assistants for literature triage, phenotype extraction, adversarial review, hypothesis generation, workbook construction, quantitative summaries and manuscript drafting. Their outputs were not treated as authorship or source evidence. The author selected the research questions, directed the analyses, resolved disputed interpretations, reviewed and approved the final text, and remains responsible for every claim in the manuscript. Gene-level claims were tied to the primary literature through the source-lock register.

Data availability

The earliest surviving public posting of the Sodium-Struggle Hypothesis is Zenodo record 19700479, version v5, published 19 October 2025: https://zenodo.org/records/19700479.

The analytical package accompanying this manuscript comprises Supplementary Data S1, Bell_Satterstrom102_v46.20_Publication_Tables_READER.xlsx (SHA-256 fb77ae6770c5894ac9be962d58f4388a4fa50224684217e5fb79cfeeaa240a4c); Supplementary Data S2, Bell_Satterstrom102_v46.20_Frozen_Full_Audit_Master_PUBLIC.xlsx (SHA-256 4e92c438a78dfc0177adb5e6a798d75fcb8b2679d68232fcf9d75bc5b7dc5a50); and Supplementary Data S3, Bell_Satterstrom102_v46.20_Public_Supplement_QA_and_Checksums.txt [10-12].

The original frozen analytical workbook has SHA-256 f8b4c3b64aa13d6fbd48a5e97265dd741ca87974b2fab05b596cc2bc29cfe551. The public spreadsheets are presentation-only derivatives of the frozen source; no analytical value, formula, status, strict count, primary domain or numerator gate was intentionally changed. The freeze identifier is NF-2026-08-11-v46.20.

References

Tables 4-6 cite the primary publications supporting the measured exemplars. The complete gene-level source-lock record for all counted endpoints is preserved in Supplementary Data S2 [11].

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